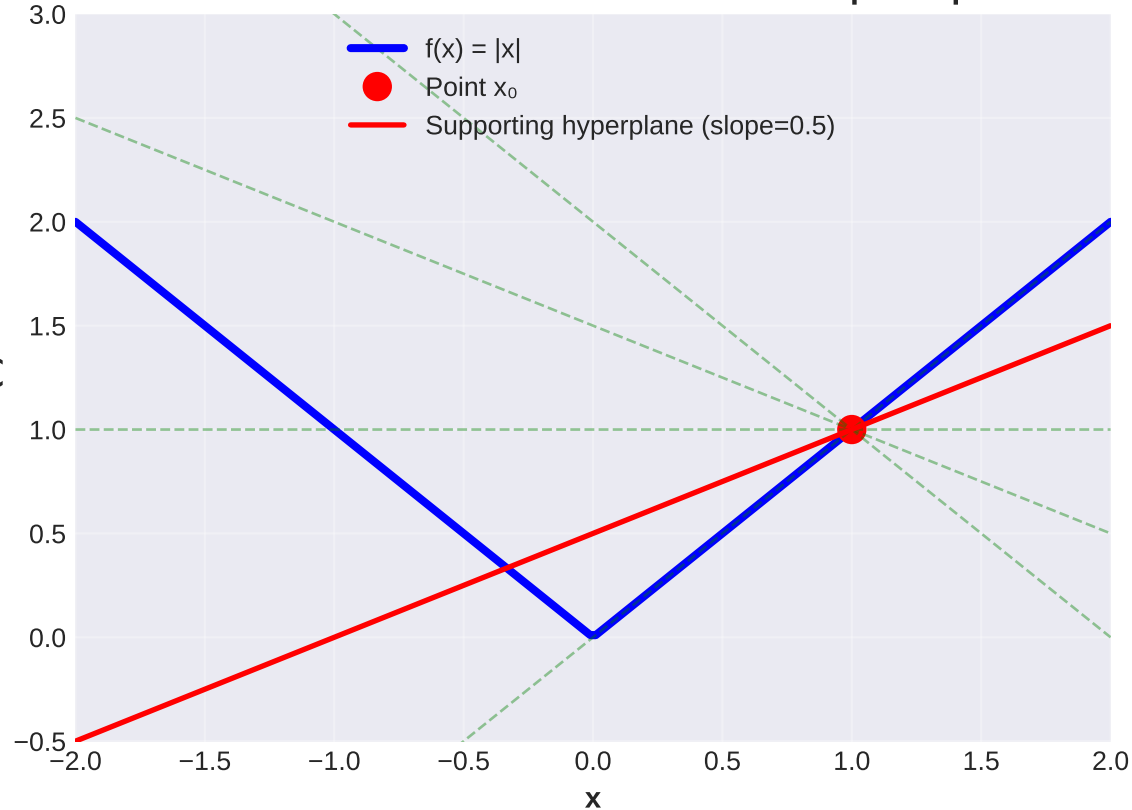


Subdifferential: Supporting Hyperplanes to Convex Functions

Non-smooth Point: Subdifferential has multiple slopes



Subdifferential Definition

At a point x_0 , the subdifferential $\partial f(x_0)$ is the set of all vectors x^* such that:

$$f(x) \geq f(x_0) + \langle x^*, x - x_0 \rangle \quad \forall x \in X$$

Key Properties:

- For smooth functions: $\partial f(x) = \{f'(x)\}$ (singleton)
- For non-smooth functions: $\partial f(x)$ is a set
- For convex functions: always non-empty
- Generalization of gradient to non-smooth functions

Example: $f(x) = |x|$ at $x_0 = 0$

$$\partial f(0) = \{z : |z| \leq 1\} = [-1, 1]$$